

Civic Voice — Grievance Reporting System

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Abstract

Public complaint redressal is an essential part of good governance. The conventional model of complaint handling involves manual processes, multiple channels of communication, and paperwork, which can lead to inefficiency and delay. This undermines trust in public institutions and impairs efficient governance. This paper introduces CivicVoice – an automated complaint redressal system utilizing artificial intelligence.

The suggested model combines the techniques of NLP, ML, and CV for analysis of both text and image data submitted by users. The CivicVoice model uses a modular multilayer design that includes four layers – presentation, application, AI processing, and database layers – which provide scalability and adaptability. Moreover, the system has real-time tracking and notification features that ensure high transparency and user engagement. The results of the experiments show that the model is highly effective in comparison with existing models; it increases classification accuracy and decreases the time of the process.

Experimentation proves an increase in response time, accuracy in classification, and efficiency compared to conventional systems. The proposed approach increases transparency, lowers manual effort, and facilitates data-driven decision-making in public management.

Keywords: Public Grievance System, Artificial Intelligence, NLP, CNN, Smart Governance, Complaint Management

I. INTRODUCTION

Management of grievances by the general public is a fundamental part of contemporary governance. Such processes enable people to submit any kind of problems associated with infrastructural facilities, utilities, and other aspects of public services. Complaints submitted manually are currently being sorted out across several hierarchical levels, causing a delay in the process as well as a lack of traceability on behalf of citizens. All of this emphasizes the need for a more efficient system for the submission and processing of citizen grievances.

Along with the development of IT technologies, as well as smart cities projects, the majority of

governments have implemented websites designed specifically for complaint submission purposes. However, most of these sites only allow users to lodge complaints while providing no automation regarding complaint classification and subsequent sorting process management.

Solutions for the above problems can be found in Artificial Intelligence. With the help of natural language processing, the system will be able to comprehend any written complaint, whereas machine learning algorithms will enable prioritizing these complaints according to context. Besides, image analysis technology can be used to evaluate any images sent by citizens as proof of their complaints. All these technologies are planned to be combined in a CivicVoice system.

The main objectives of this study are:

To design a scalable web-based platform for public grievance management.

To integrate AI modules for automatic complaint classification and prioritization.

To provide real-time complaint tracking and transparency for citizens.

To improve decision-making for government departments through analytics and data insights.

II. RELATED WORK

The development of complaint management systems has evolved significantly from traditional rule-based approaches to intelligent AI-driven systems. Early systems relied on keyword matching techniques for complaint classification, which offered limited flexibility and struggled with handling linguistic variations and contextual understanding [1], [4].

Subsequently, machine learning techniques such as Support Vector Machines, Decision Trees, and Naïve Bayes classifiers were introduced to improve classification accuracy. These approaches enhanced performance compared to rule-based systems but were highly dependent on structured datasets and required extensive feature engineering [3], [6].

Recent advancements in artificial intelligence, particularly deep learning and transformer-based models, have significantly improved text classification capabilities. These models capture semantic relationships within textual data, enabling more accurate and context-aware complaint analysis. Research has demonstrated the

effectiveness of NLP and large language models (LLMs) in automating complaint classification and routing tasks [2], [8], [14], [18].

In addition, emerging approaches focus on proactive and intelligent complaint management using semantic pattern recognition and AI-driven frameworks, which enhance efficiency in public sector informatics [7], [17]. Multimodal AI systems, capable of processing both textual and visual data, are also gaining traction for handling real-world civic issues, such as infrastructure inspection and image-based complaint analysis [5], [19].

Despite these advancements, many existing systems operate in isolation and lack seamless integration across departments, limiting their practical effectiveness. Studies have also highlighted the importance of citizen-centric AI systems for improving grievance redressal mechanisms and enhancing user experience [20], [21].

CivicVoice addresses these limitations by integrating multiple AI technologies into a unified platform that supports multimodal data processing, automated routing, and real-time tracking of complaints, thereby providing a comprehensive and scalable solution.

III. SYSTEM ARCHITECTURE

Civic Voice uses a hierarchical architecture, composed of four major layers: presentation layer, application layer, artificial intelligence layer, and database layer. The different layers serve distinct roles in the overall architecture of the system.

1. Presentation Layer

The presentation layer refers to the interface through which citizens interact with the system. The users can use the layer to register and login to the system, raise complaints, and monitor complaints' progress. It was created using React.js and Bootstrap to ensure that the design is responsive.

2. Application Layer

The application layer manages the operations of the application itself. The layer processes request from users, validates data entered by the users, and interfaces with the other components of the system. The application layer uses Node.js with Express.js to expose RESTful APIs for the complaint management process.

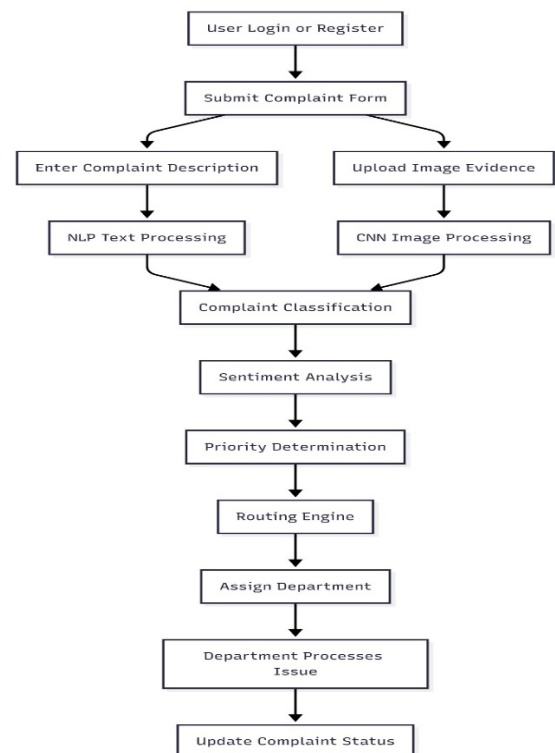
3. AI Processing Layer

AI processing layer is used to process the text and images associated with the complaint. It is comprised of several components:

- Image classification component - This is done through CNN models such as MobileNet or ResNet. These models help determine the kind of problem associated with the uploaded image.
- NLP text tagging component - This is done by applying NLP methods to determine important terms associated with complaints.
- Sentiment analysis component - Urgency level is calculated using sentiment analysis algorithms on complaint texts.
- Routing engine - Complaints are routed using output from other AI components and sent to the relevant department.

4. Data Storage Layer

Data storage layer is used to store user information, complaints, departments, and complaint updates. The database selected for this project is MongoDB due to its flexible document-oriented storage capabilities.



IV. SYSTEM IMPLEMENTATION

The implementation of the Civic Voice system follows a structured approach combining full-stack web development with artificial intelligence modules.

1. Frontend Implementation

The frontend is developed using React.js and Bootstrap to create a responsive user interface. Users can perform the following actions:

- Register and authenticate using secure login credentials
- Submit complaints with text descriptions and image attachments
- Track complaint status using a unique tracking ID
- View notifications and updates regarding complaint resolution

2. Backend Implementation

The backend is implemented using Node.js with Express.js, which manages the application logic and API communication. The backend performs tasks such as:

- Handling user authentication using JWT tokens.
- Processing complaint submissions.
- Communicating with AI modules for classification.
- Managing complaint routing and updates.

3. Database Implementation

The system uses MongoDB as the primary database. It stores collections including:

- User information
- Complaint details
- Complaint status updates
- Admin details

MongoDB enables flexible storage and efficient retrieval of structured and semi-structured data.

4. AI Module Integration

Artificial intelligence modules are implemented using Python. The following techniques are used:

A. Image Classification:

Images uploaded by users are preprocessed using OpenCV, which resizes and normalizes images before feeding them to CNN models such as MobileNet or ResNet.

B. Natural Language Processing:

Complaint descriptions are processed using NLP libraries such as spaCy or HuggingFace, which extract keywords and identify the complaint category.

C. Routing Logic:

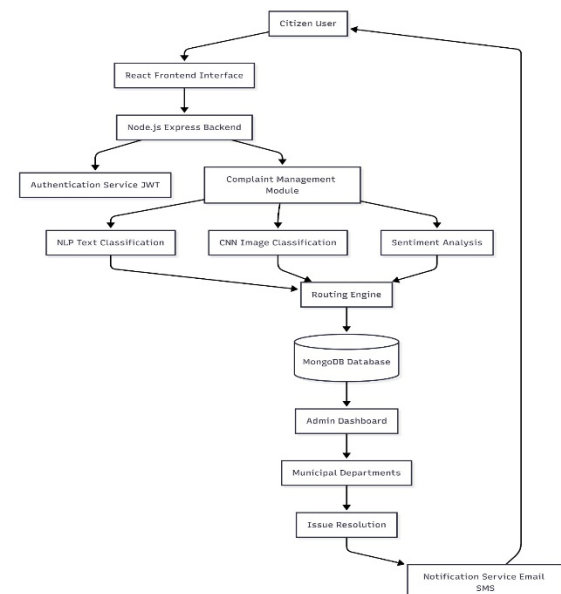
The outputs of the image and text models are combined using a weighted algorithm that determines the final complaint category and assigns it to the appropriate department.

V. DESIGN OF THE AI MODEL

The AI model in CivicVoice is made up of multiple AI components. The NLP component analyses the text content of the input complaint by breaking down the text through tokenization, removing stopwords, and extracting features from the text.

The image processing component is based on convolutional neural networks. Preprocessing procedures such as resizing and normalization enhance the effectiveness of the model. Depending on the requirements, light or deep models can be used.

Finally, a multimodal fusion strategy is employed to integrate both modalities of input. This method guarantees that the AI model makes correct choices in cases where information from one modality is ambiguous or incomplete.



VI. EXPERIMENTAL EVALUATION

The performance of CivicVoice is evaluated using multiple metrics, including classification accuracy, response time, resolution rate, and system reliability. Experimental results indicate that the system achieves high accuracy in classifying complaints and significantly reduces processing time compared to traditional methods.

Automation of classification and routing eliminates the need for manual intervention, resulting in faster response times. Real-time tracking and notification features enhance transparency and improve user satisfaction. The system also demonstrates scalability, handling multiple complaints simultaneously without performance degradation.

A. Result

```
{
  _id: ObjectId('69b44afe3511785cfda9a085'),
  description: 'pothole on the road filled with water',
  image: 'uploads/1773423353715-sample.jpg',
  category: 'Water Leak',
  department: 'Water Department',
  ai: {
    category: 'Water Leak',
    department: 'Water Department',
    urgency: 'Medium',
    priorityScore: NumberInt('55'),
    confidence: Double('0.7710134387016296'),
    source: 'image',
    keywords: [
      'pothole',
      'water'
    ]
  },
  aiMeta: {
    processingTimeSeconds: Double('4.468'),
    imageProvided: true
  },
  location: {
    lat: Double('28.641344572610116'),
    lng: Double('77.3671817779541')
  },
  status: 'pending',
  user: ObjectId('69b3ea8d28d89545ba9ad0d5'),
  createdAt: ISODate('2026-03-13T17:35:58.250Z'),
  updatedAt: ISODate('2026-03-13T17:35:58.250Z'),
  __v: NumberInt('0')
}
```

B. Performance Analysis

Compared to traditional systems:

Feature	Traditional	CivicVoice
Processing Speed	Slow	Fast
Classification	Manual	Automated
Transparency	Low	High
Scalability	Limited	High

VII. DISCUSSION

The Civic Voice system was tested for multiple criteria such as classification accuracy, response time, and system reliability.

- Classification accuracy

Since the model uses both NLP and CNN technologies, it accurately identifies the category of complaints with great precision. The use of multimodal input increases the accuracy of classification.

- Response time

The automatic assignment process greatly decreases the time it takes for the complaint to be assigned to the responsible department. Complaints are automatically processed and assigned to the relevant department immediately upon receipt.

- Transparency and citizen engagement

The application sends real-time alerts to users regarding the progress of their complaints. It increases transparency and improves the relationship between the authorities and citizens.

- Data insights

The system captures the data concerning complaint frequency, location, and category. The information can be used by the authorities to identify problems and implement preventative maintenance measures.

VIII. FUTURE WORK

Although the Civic Voice system successfully automates grievance management, several improvements can be implemented in the future.

- Development of a mobile application for Android and iOS to enable easier complaint submission.
- Integration of multilingual support and voice input, allowing users to submit complaints in regional languages.
- Integration with IoT sensors to automatically detect infrastructure issues such as water leaks or faulty streetlights.
- Use of advanced deep learning models to further improve complaint classification accuracy.

IX. CONCLUSION

In this paper, Civic Voice is described as an automated complaint management system that

employs advanced technology such as artificial intelligence techniques including NLP, image classification, and sentiment analysis to manage grievances in smart cities more effectively. Using full-stack web technologies in conjunction with AI methods, the system automates various complaint handling procedures by enabling the automatic identification and classification of complaints.

The system makes it easy for citizens to lodge and monitor complaints while allowing government departments to gain additional benefits like automatic routing and analytics. The results obtained through the experimentation have shown the potential of this technology to revolutionize complaint management systems through faster service delivery. With further improvements like application development and IoT integration, the system can be transformed into a smart governance system.

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